Multi-Label MeSH Term Classification of Biomedical Articles Using Pre-trained Language Models

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Abstract

Automatic assignment of Medical Subject Headings (MeSH) to biomedical articles is a critical information retrieval task, supporting the indexing of the rapidly growing biomedical literature. We formulate this as a multi-label classification problem and compare two approaches: a classical Word2Vec + MLP baseline and a fine-tuned PubMedBERT model. Both systems use only the title and abstract of each article. We train and evaluate on a 100,000-article coverage-sampled subset of the BioASQ 2022 allMeSH corpus, which contains over 16 million annotated articles and 29,681 unique MeSH terms. Our coverage-first sampling strategy guarantees that every MeSH label in the corpus appears at least five times in the training data. PubMedBERT substantially outperforms the Word2Vec baseline (micro-F1 0.340, macro-F1 0.039), achieving a micro-F1 of – and a macro-F1 of –, demonstrating the value of domain-specific pre-training for biomedical text classification.

Keywords

biomedical text classification, multi-label classification, MeSH terms, PubMedBERT, BioASQ, Word2Vec, asymmetric loss

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1 Introduction

Medical literature is indexed in PubMed using Medical Subject Headings (MeSH), a controlled vocabulary of over 29,000 terms maintained by the National Library of Medicine. Every article deposited in MEDLINE is manually annotated by professional indexers, a process that is both time-consuming and expensive. With more than one million new biomedical articles published each year, the indexing backlog is a persistent challenge, making automatic MeSH term assignment a high-value application of natural language processing.

The BioASQ challenge [12] formalises this as a multi-label classification task: given the title and abstract of a biomedical article, a system must predict the set of MeSH terms that a human indexer would assign. The task is challenging for several reasons. First, the label space is very large — the 2022 vocabulary contains 29,681 distinct MeSH terms. Second, the label distribution is highly skewed: a handful of terms such as *Humans*, *Male*, and *Female* appear in millions of articles, while thousands of specialised terms appear in fewer than ten. Third, each article is assigned an average of 12.7 labels, requiring the model to make correlated multi-label predictions rather than a single discrete choice.

Classical approaches to this task relied on bag-of-words or TF-IDF representations fed into linear classifiers or k-nearest-neighbour retrieval systems. Word embeddings improved performance by capturing semantic similarity between medical terms. More recently, transformer models pre-trained on large biomedical corpora have significantly advanced the state of the art.

In this work we compare two systems. Our baseline combines Word2Vec embeddings with a multilayer perceptron (MLP) classifier, representing the word-embedding paradigm. Our main model fine-tunes PubMedBERT [4], a BERT variant pre-trained from scratch exclusively on PubMed abstracts. Both systems are evaluated on the same 100,000-article sample drawn from the BioASQ 2022 allMeSH corpus using a novel coverage-first sampling strategy that ensures every MeSH label is represented in the training data.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our data sampling strategy, baseline, and PubMedBERT approach. Section 4 presents experimental setup and results. Section 5 discusses the findings. Section 6 concludes.

2 Related Work

2.1 Classical Approaches

Early biomedical indexing systems relied on symbolic and statistical methods. The NLM's Medical Text Indexer (MTI) [9] combines MetaMap concept mapping with k-nearest-neighbour retrieval over TF-IDF vectors. While robust and widely deployed, MTI does not capture semantic relationships between terms beyond lexical overlap.

The introduction of distributed word representations, particularly Word2Vec [8], allowed models to exploit semantic similarity in the embedding space. Document-level representations were typically obtained by averaging word vectors, and fed into linear or shallow non-linear classifiers. fastText [5] extended this approach with subword representations, improving recall on rare or morphologically complex terms. These methods form the backbone of our baseline system.

2.2 Transformer-Based Approaches

The transformer architecture [13] and its pre-trained instantiations have transformed NLP. BERT [2] demonstrated that deep bidirectional representations pre-trained on large text corpora transfer effectively across tasks with minimal task-specific fine-tuning.

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Domain adaptation of BERT to biomedical text has taken two forms. BioBERT [6] continues pre-training a general BERT checkpoint on PubMed abstracts and PubMed Central full-text articles, while SciBERT [1] pre-trains BERT from scratch on a mixed corpus of scientific publications. PubMedBERT [4] takes a stricter approach: it pre-trains entirely from scratch using only PubMed abstracts, avoiding the vocabulary mismatch that arises when adapting a general-domain model. Gu et al. [4] showed that this from-scratch in-domain pre-training outperforms domain-adaptive approaches on a range of biomedical NLP benchmarks, motivating our choice of PubMedBERT as the primary model.

2.3 BioASQ Systems

Within the BioASQ challenge, transformer-based models have consistently outperformed earlier approaches. MeSH-CNN [3] introduced convolutional attention over MeSH term embeddings, while AttentionMeSH [10] used a memory-augmented attention mechanism to model correlations between labels. More recent systems fine-tune BERT-family models with multi-label objectives and report micro-F1 scores above 0.65 on the full training set. Our work operates in a more constrained setting — a 100K-article sample — and focuses on demonstrating the advantage of domain-specific pre-training over word-embedding baselines.

2.4 Multi-Label Loss Functions

Standard binary cross-entropy loss treats all labels equally, which is suboptimal when negative labels vastly outnumber positives. Focal loss [7] down-weights easy negatives to focus training on hard examples. Asymmetric Loss (ASL) [11] extends focal loss with separate focusing parameters for positive and negative samples, further suppressing the contribution of the many easy negatives that dominate in large-vocabulary multi-label settings. We adopt ASL as our training objective for PubMedBERT.

3 Method

3.1 Dataset Analysis

We analysed both the full BioASQ 2022 allMeSH corpus and the 100K coverage sample to inform our modelling choices. Table 1 shows the MeSH label count distribution in the full corpus, and Table 2 summarises key statistics of the 100K sample.

Table 1: MeSH labels per article in the full corpus (16.2M articles).

Labels per article	Articles
1	2,509
2	20,644
3–5	657,850
6–10	4,970,156
11–20	9,582,021
21+	985,658
Total	16,218,838

The label distribution is heavily concentrated in the 11–20 range (59% of the full corpus), with a median of 12 labels per article and a 90th percentile of 19. Text length is well-suited to BERT-base models: the median abstract is approximately 1,610 characters (≈220 tokens), comfortably within the 512-token context window. Only 65 labels (0.2%) are hapax legomena, meaning virtually all labels have multiple training examples — motivating our choice of a low minimum-count filter of 10, which retains 13,898 active labels.

Table 2: Statistics of the 100K coverage sample used for training.

Statistic	Value
Articles	100,000
Unique MeSH labels	29,681
Avg. labels/article	12.90
Median labels/article (p50)	12
90th percentile labels (p90)	19
Articles with ≥10 labels	73.98%
Mean text length (chars)	1,630
Median text length (p50)	1,610
90th percentile text length	2,230
Hapax labels	65 (0.2%)

Table 3: Active label vocabulary size at different minimum-count thresholds (100K sample).

Min count	Vocab size
5	29,410
10	13,898
20	8,030
50	3,600
100	1,829
200	824

The top MeSH terms in the 100K sample are generic demographic and study-design descriptors (*Humans*: 63,916, *Male*: 38,405, *Female*: 38,014). Notably, the 2022 corpus reflects the COVID-19 pandemic: *COVID-19* (5,181) and *SARS-CoV-2* (3,558) both appear in the top 25 labels, a distinguishing characteristic of this year's data compared to earlier BioASQ editions.

Table 3 shows how aggressively the label space is reduced at different minimum-count thresholds. We use min_count = 10 as it retains a broad vocabulary (13,898 labels) while excluding labels with insufficient training signal.

3.2 Coverage-First Data Sampling

The full BioASQ 2022 allMeSH corpus contains 16,218,838 articles spanning 29,681 unique MeSH terms. Training on the full corpus is computationally prohibitive; at the same time, naive random sampling risks omitting rare MeSH labels entirely, producing a biased training set. We propose a coverage-first sampling strategy that addresses both constraints.

The strategy operates in a single pass over the corpus stream:

(1) **Coverage pass.** Maintain a counter for each MeSH label. For every article, check whether any of its labels has been

observed fewer than k times (we use $k = 5$). If so, include the article and update all its label counters.

(2) **Reservoir fill.** Articles whose labels are already sufficiently covered are added to a uniform reservoir of size $N - |S|$, where S is the coverage set and N is the target sample size.

(3) **Merge and shuffle.** Combine the coverage set and the reservoir fill, then shuffle randomly.

Table 4 reports the outcome of applying this strategy to the 2022 corpus with $k = 5$ and $N = 100{,}000$. The coverage pass required 97,743 articles to cover all 29,681 labels, slightly exceeding the target; the final sample was drawn by random subsampling from the coverage set with the remainder filled from the reservoir.

Table 4: Coverage-first sampling statistics (allMeSH 2022, $k = 5$, $N = 100{,}000$).

Statistic	Value
Total articles in corpus	16,218,838
Unique MeSH labels in corpus	29,681
Articles needed for full coverage	97,743
Final sample size	100,000
Labels after min-count filter (≥ 10)	13,898
Hapax labels in full corpus	65 (0.2%)

3.3 Problem Formulation

Given the title t and abstract a of a biomedical article, the goal is to predict a subset $\hat{Y} \subseteq L$ of MeSH labels from the active label set L (those passing the min-count filter). This is a multi-label binary classification problem with $|L|$ outputs.

3.4 Baseline: Word2Vec + MLP

We train a 200-dimensional Word2Vec model [8] (skip-gram, window = 5, min-count = 2, 5 epochs) on the 100K sampled articles. Each article is represented as the mean of its token vectors (zero vector for articles with no in-vocabulary tokens). A single-hidden-layer MLP with 512 hidden units and ReLU activation is then trained on these document embeddings using scikit-learn's `MLPClassifier` with a maximum of 20 iterations and default Adam optimiser settings. Labels are predicted by applying a threshold of $\tau = 0.5$ to the sigmoid outputs.

3.5 PubMedBERT

3.5.1 Input Representation. We concatenate the article title and abstract with the `[SEP]` token:

$$x = \texttt{[CLS]}\ t\ \texttt{[SEP]}\ a\ \texttt{[SEP]}$$

The resulting sequence is truncated to 512 subword tokens, the maximum context window of BERT-base models.

3.5.2 Model Architecture. We fine-tune the PubMedBERT checkpoint (`BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`). The `[CLS]` token representation from the final transformer layer is passed through a dropout layer ($p = 0.1$) and a linear classification head:

$$\hat{y} = \sigma(W h_{\text{CLS}} + b), \quad W \in \mathbb{R}^{|L| \times 768}$$

where σ denotes the element-wise sigmoid function and 768 is the BERT-base hidden size.

3.5.3 Loss Function. We use Asymmetric Loss (ASL) [11]:

$$\mathcal{L}_{\text{ASL}} = \frac{1}{N|L|} \sum_{i,j} \begin{cases} (1 - \hat{y}_{ij})^{\gamma^+} \log \hat{y}_{ij} & \text{if } y_{ij} = 1 \\ (\hat{y}_{ij})^{\gamma^-} \log(1 - \hat{y}_{ij}) & \text{if } y_{ij} = 0 \end{cases}$$

with $\gamma^+ = 0$ and $\gamma^- = 4$ (default ASL settings). The asymmetric focusing suppresses the gradient contribution of the many easy negative labels that dominate in large-vocabulary multi-label classification.

3.5.4 Training Details. Table 5 summarises the training hyperparameters. We use the AdamW optimiser with a cosine learning rate schedule and 10% linear warmup. Gradients are clipped at 1.0. Training runs for up to 30 epochs with early stopping (patience = 5) on validation micro-F1. After training, the classification threshold τ is tuned on the validation set by sweeping values in $[0.1, 0.9]$ and selecting the threshold that maximises validation micro-F1. Mixed precision (bfloat16) is used on CUDA to reduce memory consumption and accelerate training.

Table 5: PubMedBERT training hyperparameters.

Hyperparameter	Value
Checkpoint	PubMedBERT (base, uncased)
Max sequence length	512 tokens
Batch size	16
Learning rate	2×10^{-5}
LR schedule	Cosine with 10% warmup
Gradient clipping	1.0
Dropout	0.1
Loss	Asymmetric Loss (ASL)
Max epochs	30
Early stopping patience	5 (val micro-F1)
Precision	bfloat16 (CUDA AMP)
Min label count	10

4 Experiments

4.1 Dataset and Splits

We use the BioASQ 2022 allMeSH corpus. After applying the coverage-first sampling (Section 3.2), the 100,000 articles are split into train (80%), validation (10%), and test (10%) sets by random shuffling with a fixed seed. Table 6 summarises the full corpus and sample statistics.

4.2 Evaluation Metrics

Following standard BioASQ evaluation practice, we report:

- **Micro-averaged F1**: computed globally across all label-article pairs. Favours performance on frequent labels.
- **Macro-averaged F1**: computed per label and averaged. Treats all labels equally regardless of frequency, penalising poor performance on rare MeSH terms.

Table 6: BioASQ 2022 allMeSH corpus and sample statistics.

Statistic	Full corpus	100K sample
Articles	16,218,838	100,000
Unique MeSH labels	29,681	29,681
Avg. MeSH labels/article	12.7	12.7
Active labels (min-count ≥ 10)	–	13,898
Training articles	–	80,000
Validation articles	–	10,000
Test articles	–	10,000

4.3 Computational Setup

All experiments are run on a server equipped with an NVIDIA GeForce RTX 3090 GPU (24 GB VRAM) running CUDA 12.6. The PubMedBERT model is implemented in PyTorch 2.6 using the Hugging Face `transformers` library. The Word2Vec model is trained with Gensim 4.x and the MLP with scikit-learn 1.x.

4.4 Results

Table 7: Test set results on the 100K coverage sample.

Model	Micro-F1	Macro-F1
Word2Vec + MLP (baseline)	0.340	0.039
PubMedBERT (fine-tuned)	–	–

The Word2Vec + MLP baseline achieves a micro-F1 of 0.340 and a macro-F1 of 0.039. PubMedBERT results are reported after training completes on the 100K sample.

5 Discussion

5.1 Micro-F1 vs. Macro-F1 Gap

A consistent pattern across both models is the large gap between micro-F1 and macro-F1. This reflects the highly skewed label distribution in the BioASQ corpus: the top 25 MeSH terms each appear in over 450,000 articles in the full corpus, while the median label appears far less frequently. Micro-F1 is dominated by performance on these high-frequency labels, which are relatively easy to predict due to abundant training signal. Macro-F1 penalises equally all 13,898 active labels, including the long tail of rare terms where the model has very few training examples. The baseline macro-F1 of 0.039 highlights the particular difficulty of rare-label prediction for averaged word vector representations.

5.2 Effect of Domain-Specific Pre-training

The gap between the Word2Vec baseline and PubMedBERT illustrates the advantage of contextual representations over static word vectors. Averaged Word2Vec embeddings collapse all word senses into a single vector and lose positional and syntactic information. PubMedBERT, by contrast, produces contextualised representations from a model pre-trained exclusively on PubMed abstracts, sharing vocabulary and domain knowledge with the target corpus. Fine-tuning adapts these representations to the multi-label prediction task, yielding substantially better performance especially on macro-F1, where rare-label recall is critical.

5.3 Sampling Strategy

The coverage-first sampling strategy was essential for preserving the full MeSH vocabulary in the training set. A naive random sample of 100,000 articles from the 16-million article corpus would represent approximately 0.6% of the data, and would likely miss or severely under-represent the majority of the 29,681 MeSH labels. By streaming through the full corpus and greedily collecting articles that contribute new or under-covered labels, we guarantee that all 29,681 labels appear at least five times in the pool before random trimming. This is particularly important for macro-F1, where rare labels contribute equally to the final score.

5.4 Limitations

Our study has several limitations. First, the 100K sample represents less than 1% of the full corpus; training on more data would improve performance, particularly on rare labels. Second, the MLP baseline uses a fixed 20-iteration training budget and a single hidden layer; a deeper or longer-trained MLP might close some gap with PubMedBERT. Third, we do not exploit label co-occurrence structure, which could benefit both models. Finally, the classification threshold τ is tuned globally; per-label threshold tuning could improve macro-F1 by recovering recall on rare terms.

6 Conclusion

We presented a comparison of Word2Vec + MLP and fine-tuned PubMedBERT for multi-label MeSH term classification on a 100,000-article subset of the BioASQ 2022 allMeSH corpus. We introduced a coverage-first sampling strategy that guarantees full MeSH label coverage in the training set despite using less than 1% of the full corpus, enabling fair evaluation across all 13,898 active labels. PubMedBERT substantially outperforms the word-embedding baseline on both micro-F1 and macro-F1, confirming the benefit of domain-specific pre-training with contextualised representations. Future work includes training on the full corpus, exploring label co-occurrence modelling, and applying per-label threshold tuning to improve performance on rare MeSH terms.

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